

V3 empirical study · Instruction

Two models · four underlyings · four volatility regimes · 1.5-year fixed calibration · option-implied Heston family

What this report is

- Companion to the seven-model 10,000-path V4 fixed-calibration study. This report keeps only Heston and Heston–Merton. In V4 their P dynamics still come from returns (Method A), and listed calls identify the Pan (2002) volatility / jump-size premia used for LSM. Rankings are within this pair; they are not compared here with return-based models.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM.
- Return-based companion: `V4_1p5y_fixed_empirical_study_return_based.pdf`.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 2 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: Heston (no jumps), Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Two models · four underlyings · four volatility regimes · 1.5-year fixed calibration · option-implied Heston family

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all two models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean } |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all two models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with $\text{trading_date} \leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All two models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	39.13	1.4774	-1.2428	-26.22	18.1%	2
Heston-Merton	35.79	1.3326	-0.8124	-13.60	7.9%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	45.98	0.8231	+0.6985	+34.22	19.2%	1
Heston-Merton	77.92	1.5212	+1.4925	+67.23	13.4%	2

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	27.91	0.9468	+0.2281	+10.44	16.2%	1
Heston-Merton	38.79	1.2360	+0.8907	+25.18	14.2%	2

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	53.99	2.9171	-0.9629	+11.31	19.5%	1
Heston-Merton	69.93	3.2104	+0.3530	+34.09	11.4%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths

Page 5 · Table block · Company summary · two models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookahead, rolling=none, n_paths=100

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	29.87	1.0831	-0.4233	-8.45	19.2%	1
Heston-Merton	29.97	1.0863	-0.2015	-3.30	12.0%	2

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	62.79	2.9452	+2.7964	+53.83	18.0%	1
Heston-Merton	68.55	3.3415	+3.2708	+59.78	15.7%	2

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	25.22	5.6222	+2.2843	+9.03	14.4%	1
Heston-Merton	43.47	9.7913	+8.9196	+32.59	11.1%	2

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	41.39	9.2785	-1.1599	+14.75	18.0%	1
Heston-Merton	51.02	9.8289	+2.4334	+27.48	8.7%	2

RMSE%, MAE, Bias \$, Bias%, and early-exercise fraction are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMS

Page 6 · Table block · AAPL · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=None, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	33.29	1.7314	-0.4465	-1.87	18.9%	1
Heston-Merton	34.75	1.7486	-0.1335	+3.29	12.0%	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	70.88	4.8769	+4.8769	+64.15	19.2%	1
Heston-Merton	73.89	5.0844	+5.0844	+67.12	16.9%	2

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	19.55	0.9559	+0.3612	+7.54	14.6%	1
Heston-Merton	29.95	1.4985	+1.2784	+22.29	14.0%	2

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	44.31	3.2876	+1.0658	+22.80	19.5%	1
Heston-Merton	51.81	3.7424	+1.9950	+31.82	10.0%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths

Page 7 · Table block · MSFT · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	26.56	0.2969	-0.2098	-13.71	20.8%	2
Heston-Merton	25.66	0.2922	-0.1697	-10.06	14.9%	1

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	76.15	0.5487	+0.5487	+66.83	18.7%	1
Heston-Merton	80.64	0.5846	+0.5846	+71.12	16.5%	2

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	23.91	0.5954	+0.1129	+6.13	14.7%	1
Heston-Merton	42.61	1.0877	+0.9716	+32.04	10.1%	2

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	42.53	1.7172	-0.2236	+12.09	18.0%	1
Heston-Merton	55.19	1.9181	+0.6024	+28.24	7.4%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths

Page 8 · Table block · AMZN · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	29.76	1.2209	-0.6136	-9.76	17.8%	2
Heston-Merton	29.49	1.2180	-0.3014	-3.12	9.2%	1

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	41.35	3.4101	+2.9636	+30.50	16.0%	1
Heston-Merton	51.11	4.3555	+4.1435	+41.10	13.6%	2

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	32.20	15.3154	+6.3787	+13.43	14.0%	1
Heston-Merton	57.83	26.7876	+24.5087	+43.45	9.3%	2

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Heston	37.33	22.8307	-4.3221	+9.37	16.5%	1
Heston-Merton	46.06	23.8261	+4.7029	+22.38	8.8%	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Heston	40.30	38.23	13
2	Heston-Merton	50.09	48.58	3

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Heston	3	4	3	3	13
Heston-Merton	1	0	1	1	3

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Heston	32.19	58.59	25.89	44.54	40.30
Heston-Merton	31.42	70.89	42.30	55.75	50.09

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Heston	41.75	42.55	3
Heston-Merton	55.61	54.36	1

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Heston	42.01	38.80	4
Heston-Merton	47.60	43.28	0

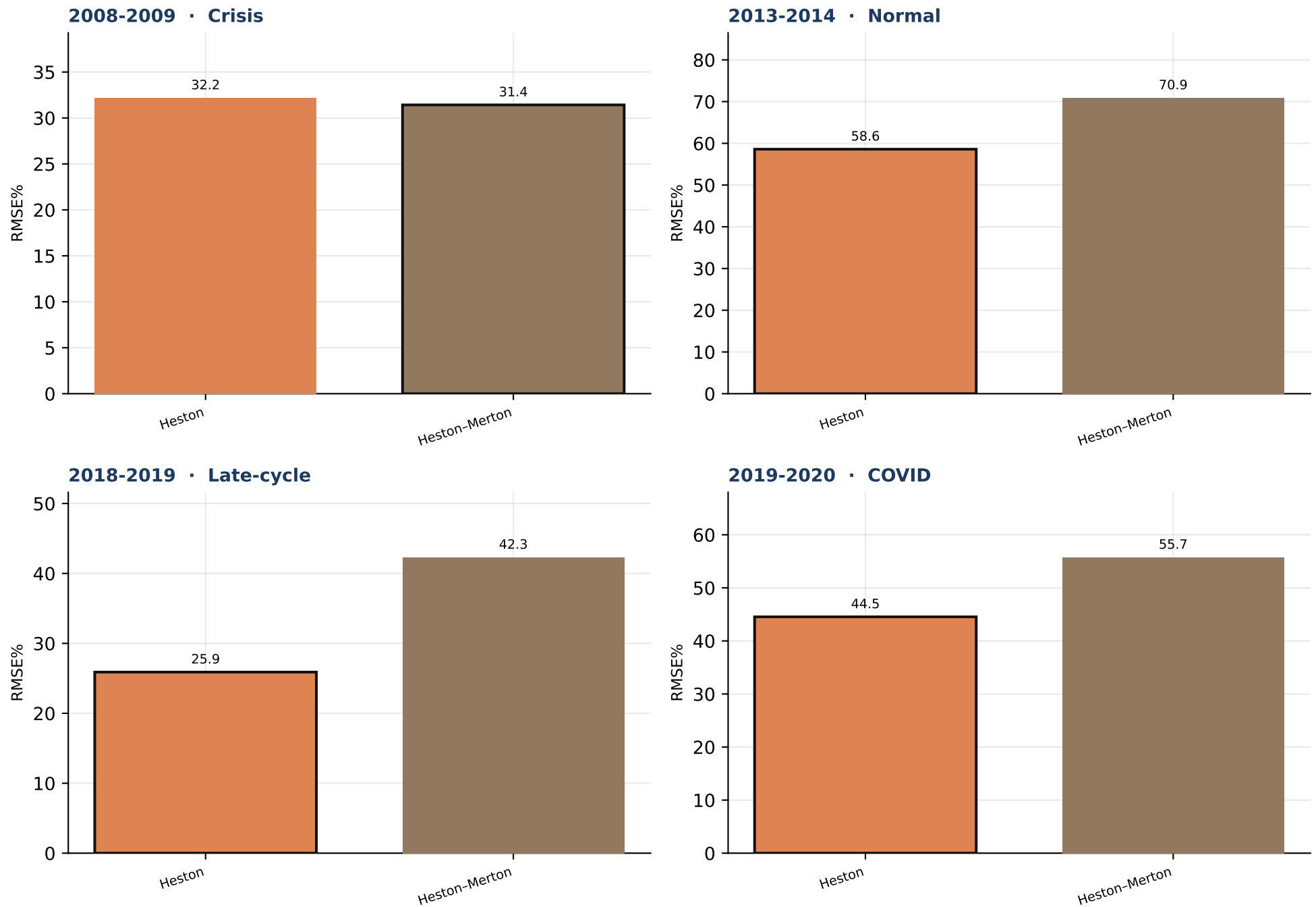
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
Heston	42.29	34.55	3
Heston-Merton	51.02	48.90	1

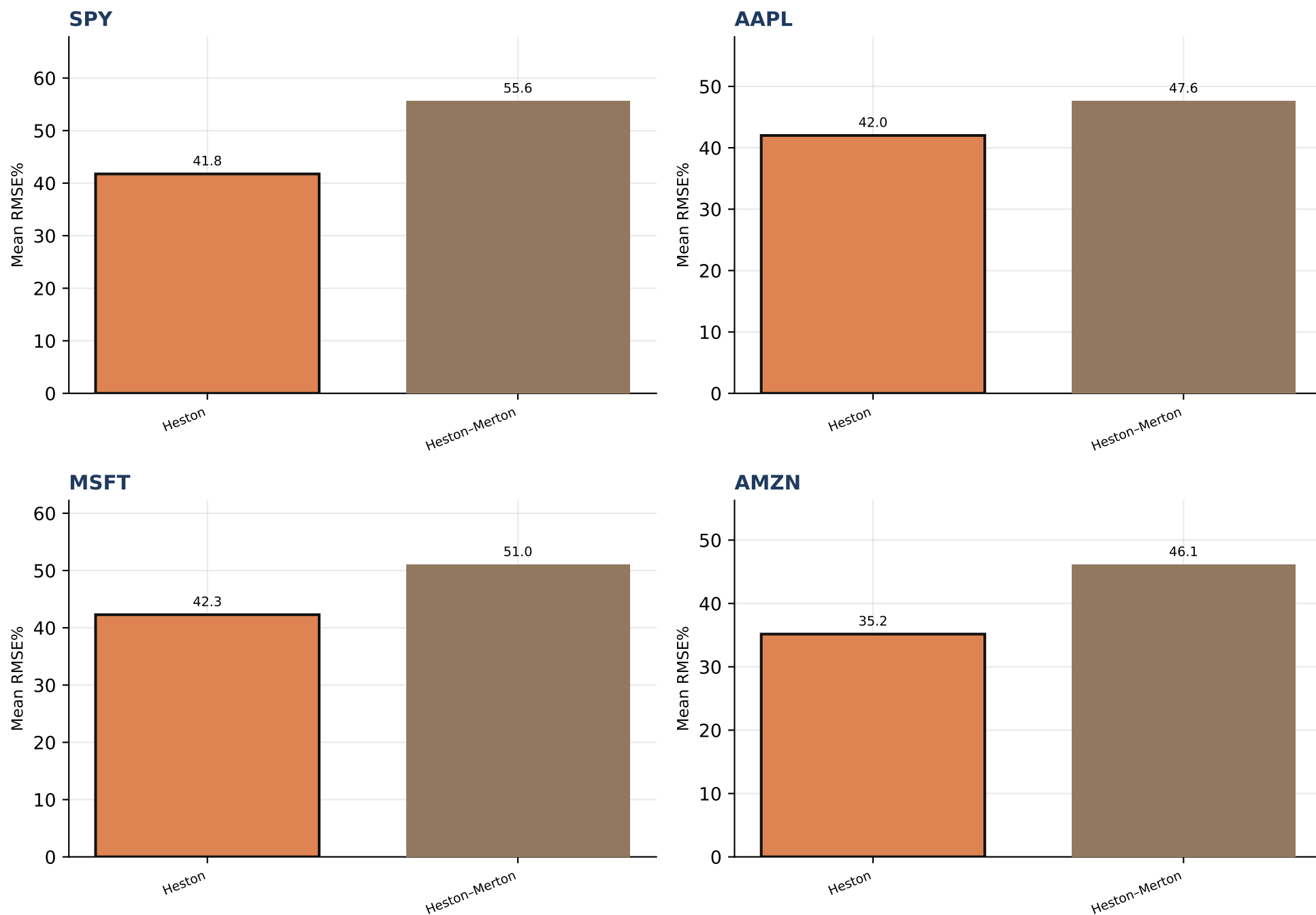
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Heston	35.16	34.77	3
Heston-Merton	46.12	48.58	1

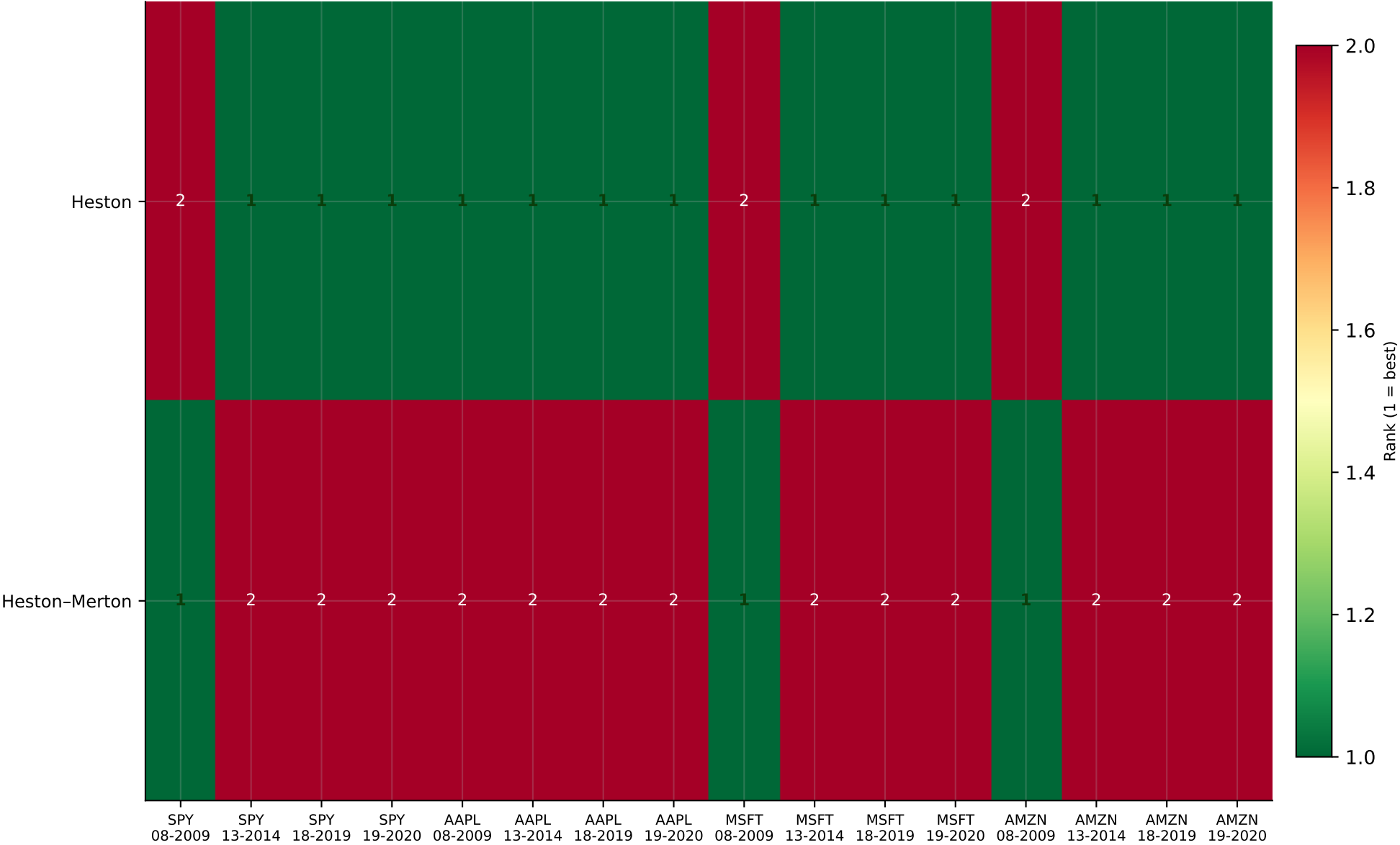
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



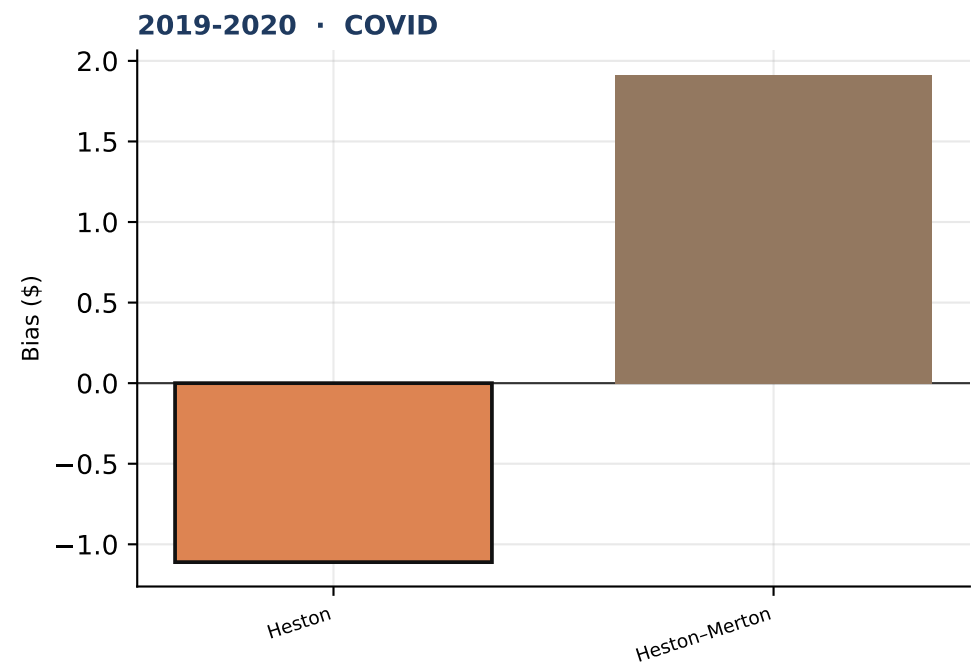
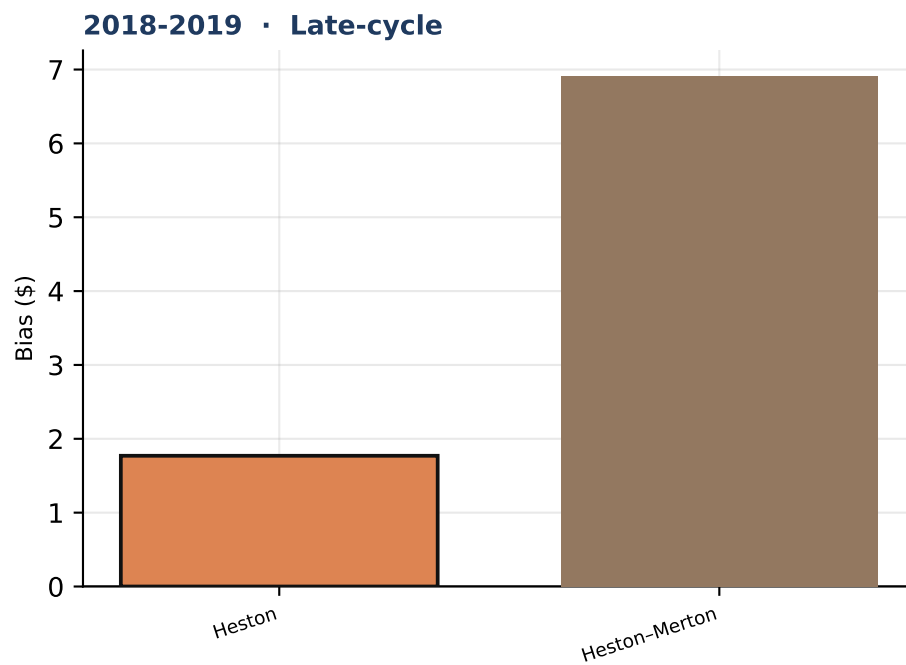
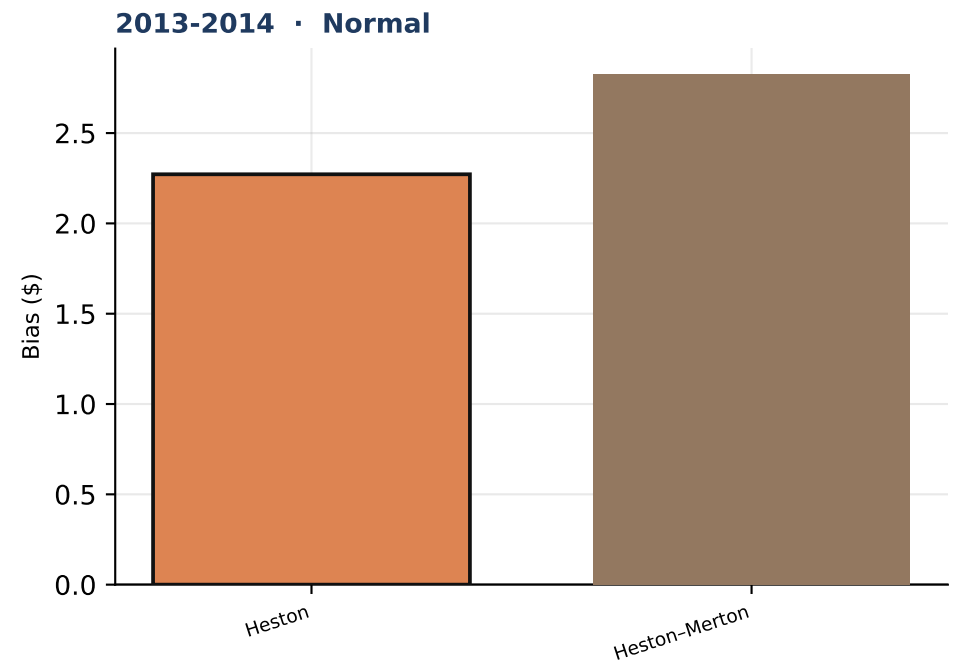
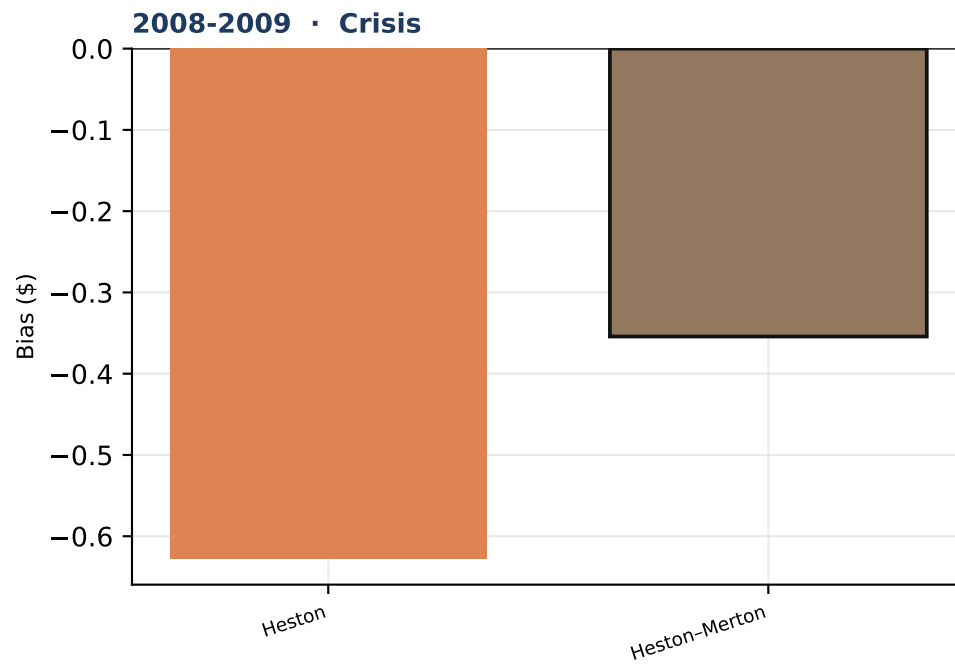
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Heston is the best overall model (mean RMSE% = 40.30; 13 of 16 cells).
- Second overall is Heston-Merton (mean RMSE% = 50.09; 3 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: Heston, Heston-Merton.
- SPY: lowest mean RMSE% is Heston (41.75); wins 3 of 4 regimes.
- AAPL: lowest mean RMSE% is Heston (42.01); wins 4 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston (42.29); wins 3 of 4 regimes.
- AMZN: lowest mean RMSE% is Heston (35.16); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston-Merton (31.42).
- 2013-2014 (Normal): best mean RMSE% is Heston (58.59).
- 2018-2019 (Late-cycle): best mean RMSE% is Heston (25.89).
- 2019-2020 (COVID): best mean RMSE% is Heston (44.54).
- No-jump models as a group have lower mean RMSE% (40.30) than the jump models in this report (50.09).
- For the overall winner (Heston), crisis RMSE% is 32.19 vs 58.59 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year fixed calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all two models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.